WORKED SOLUTION

STEP III 2004, Question 9

Mechanics • Circular Motion

STEP III 2004, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q9

The key idea

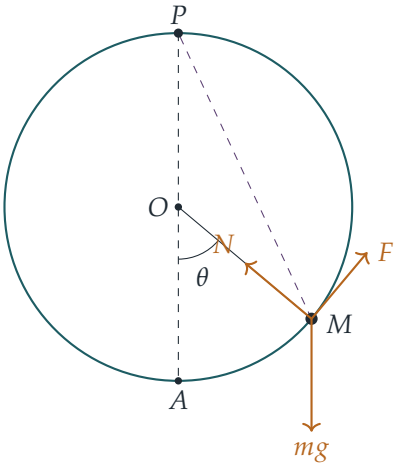
The hoop's weight acts through O, which is directly below the axis, so it has no moment about P. The only other force on the hoop (apart from the reaction at the axis) is the contact force from the mouse, so the hoop stays put exactly when that contact force has zero moment about P, that is, when its line of action passes through P. Everything follows from asking when the mouse's contact force can point along MP: once for the speed, and once for the friction limit.

Solution

The model

Treat the mouse as a particle of mass m and the hoop as a rigid ring (its mass will turn out not to matter) that turns freely about a smooth horizontal axis through P. Let $\theta = \angle AOM$, measured from the downward vertical, with the mouse running in the direction of increasing θ. While the hoop is at rest the mouse moves on a fixed circle of radius a about O at constant speed u.

The forces on the mouse are its weight mg, the normal reaction N from the hoop towards O, and friction F along the tangent, which we take as positive in the direction of increasing θ (up the hoop).



Resolving for the mouse along the tangent (no tangential acceleration, as the speed is constant) and along the radius (acceleration u^2/a towards O):

$$F = mg\sin\theta,$$
$$N - mg\cos\theta = \frac{mu^2}{a}.$$

When is the hoop in equilibrium?

By Newton's third law the mouse pushes on the hoop with a force equal and opposite to the contact force (N and F) on the mouse. The hoop's weight acts at O, vertically below P, and the axis reaction acts at P, so neither has a moment about P. The hoop therefore stays in equilibrium with POA vertical if and only if the contact force has zero moment about P, that is, its line of action passes through P.

Triangle OPM is isosceles ($OP = OM = a$) with $\angle POM = \pi - \theta$, so $\angle OMP = \frac{1}{2}\theta$. The line MP thus makes an angle $\frac{1}{2}\theta$ with the radius MO, tilted towards the direction of increasing θ. So the contact force on the mouse points along MP exactly when

$$\frac{F}{N} = \tan\tfrac{1}{2}\theta. \qquad (*)$$

(Equivalently, taking moments about P directly, the moment of the mouse's force on the hoop is $a[N\sin\theta - F(1+\cos\theta)]$, and $\sin\theta/(1+\cos\theta) = \tan\frac{1}{2}\theta$.)

? **Ask yourself** How do we turn this into a condition on u?

Use the equations of motion. The contact force on the mouse is its mass times acceleration minus its weight:

$$\mathbf{R} = m\cdot\frac{u^2}{a}\,\widehat{MO} + mg\,\widehat{\text{up}}.$$

But $\overrightarrow{MP} = \overrightarrow{MO} + \overrightarrow{OP} = a\,\widehat{MO} + a\,\widehat{\text{up}}$, with equal coefficients. So $\mathbf{R}$ is parallel to MP,

for every θ, if and only if $u^2/a = g$, that is,

$$u = \sqrt{ag}.$$

Algebraically: with $u^2 = ag$ the equations of motion give $F = mg\sin\theta$ and $N = mg(1+\cos\theta)$, so

$$\frac{F}{N} = \frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta}{2\cos^2\frac{1}{2}\theta} = \tan\tfrac{1}{2}\theta,$$

which is $(*)$. For a general speed the moment about P works out as $m\sin\theta\,(u^2 - ag)$, which vanishes at every position only when $u^2 = ag$. So the mouse can keep the hoop in equilibrium by running at speed $\sqrt{ag}$, as required, provided the hoop can actually supply this friction. We look at that next.

The friction limit

The mouse's feet grip without slipping only while $|F| \le \mu N$. With $u = \sqrt{ag}$, and on the lower part where $|\theta| < \frac{1}{2}\pi$ so that $\cos\frac{1}{2}\theta > 0$, this is

$$\tan\tfrac{1}{2}|\theta| \le \mu \quad\Longleftrightarrow\quad |\theta| \le 2\arctan\mu.$$

So everything works until $\angle AOM = 2\arctan\mu$. (This point is on the lower half of the hoop only if $\mu < 1$; if $\mu \ge 1$ the mouse never reaches it.)

Past this point no choice of speed helps. Equilibrium needs $F/N = \tan\frac{1}{2}\theta > \mu$, and friction can't give that. With any speed and any friction satisfying $|F| \le \mu N$, the moment about P of the mouse's force on the hoop is

$$a[N\sin\theta - F(1+\cos\theta)] \;\ge\; aN(1+\cos\theta)(\tan\tfrac{1}{2}\theta - \mu) > 0.$$

What happens

Once the mouse passes $\angle AOM = 2\arctan\mu$ it can no longer grip hard enough. Its feet slip, friction falls to its limiting value μN, and the mouse can't hold its speed: it begins to slide back. Its contact force on the hoop now passes to the side of P, so it has a nonzero moment, and the hoop is no longer in equilibrium. The hoop starts to rotate about P. The moment above is in the sense that the normal push of the mouse wins over the reduced friction, so the hoop swings with A moving towards the mouse's side, the way the mouse was running. From then on the motion is no longer the simple equilibrium picture: the hoop swings about P while the mouse slides on it.

Checks

- At $\theta = 0$ the mouse is at A, the contact force $N = 2mg$ is vertical and passes through P, and no friction is needed, as expected.
- The speed $\sqrt{ag}$ has units of speed, doesn't depend on μ or on the hoop's mass, and makes the normal reaction $mg(1+\cos\theta)$, positive everywhere on the lower half.

- For $\mu = 0.5$ the critical angle is $2\arctan 0.5 \approx 0.927$ rad $\approx 53.1^\circ$. There $\sin\theta = 0.8$ and $1 + \cos\theta = 1.6$, so $F/N = 0.5 = \mu$ exactly. A numerical check of the moment about P at random positions, with $u^2 = ag$, gave zero to within $10^{-15}mga$.

Common slips

- Taking moments about O instead of P. The hoop pivots about P, and the normal reaction (which passes through O) does have a moment about P.
- Forgetting the centripetal term in N, which gives $N = mg\cos\theta$ and the wrong friction condition.
- Claiming the mouse simply "falls off". On the lower half $N > 0$ throughout. What fails is the grip, not the contact.

Beyond STEP

The neat fact here is that a body is in equilibrium about a pivot exactly when the resultant of the other forces passes through the pivot. The "line of action through P" argument replaces a page of resolving. The second half is a first taste of coupled systems, where a slipping contact feeds one body's motion into another's. Analysing the motion after slipping needs the hoop's moment of inertia and the full equations of rigid-body rotation. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both torque about a fixed axis and friction-limited motion in this style.

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